MSBA BOOTCAMP · DAY 5

Git & GitHub Basics

Version control in a nutshell: track every change to your work, back it up in the cloud, and collaborate without emailing files named `final_v3_REALLY_FINAL.ipynb`.

1. What Are Git and GitHub?

Git is a program on your computer that records snapshots of a project over time, so you can see what changed, when, and by whom, and go back if needed. **GitHub** is a website that hosts Git repositories online: your backup, your portfolio, and the place where teams share code.

Repository	A project folder that Git is tracking ("repo" for short). The history lives in a hidden `.git` subfolder.
Commit	One saved snapshot of the project, with a message describing what changed. A repo's history is a chain of commits.
Stage	The waiting area for the next commit. You pick which changes go in with `git add`.
Remote	A copy of the repo hosted elsewhere, usually on GitHub. By convention it is named `origin`.
Clone	Your local copy of a remote repo, with its full history.
Push / Pull	Send your new commits up to GitHub / bring down commits that are on GitHub but not yet on your machine.

2. One-Time Setup

Tell Git who you are (this signs your commits) once per computer:

```
git config --global user.name  "Ada Lovelace"
git config --global user.email "ada@example.com"
git --version                  # confirm Git is installed
```

> **Authentication:** the first time you push to GitHub, it will ask you to sign in. Follow the browser prompt (or use a personal access token as your password). This is also a one-time setup per computer.

3. Getting a Repository

`git clone https://github.com/user/repo.git`	Download a repo from GitHub into a new folder, history included. This is how you get the course repo.
`git init`	Start tracking the current folder as a brand-new repo (when you are creating a project from scratch).

4. The Everyday Cycle

Ninety percent of Git is this loop. Edit your files, then:

`git status`	What changed? What is staged? Run this constantly; it also suggests the next command.
`git add report.ipynb`	Stage one file for the next commit. `git add .` stages everything that changed.
`git commit -m "Add revenue analysis"`	Save the snapshot with a short present-tense message describing the change.

`git push`	Upload your new commits to GitHub.
`git pull`	Download and apply commits from GitHub (do this before you start working, and to get instructor updates).

5. Looking at History and Changes

`git log`	The commit history: author, date, and message for each snapshot. Quit with `q`.
`git log --oneline`	The same, one commit per line. Much easier to scan.
`git diff`	Show exactly what you have changed since the last commit, line by line.
`git diff --staged`	Show what is staged, i.e. what the next commit will contain.

6. Undoing Things (The Safe Ones)

`git restore report.ipynb`	Throw away your uncommitted edits to a file and go back to the last committed version. (Cannot be undone; your edits are gone.)
`git restore --staged report.ipynb`	Un-stage a file you added by mistake. The edits themselves are kept.
`git commit --amend -m "Better message"`	Fix the message of the commit you just made (only if you have not pushed it yet).

7. Telling Git What to Ignore

A file named `.gitignore` in the repo's top folder lists patterns Git should never track: temporary files, secrets, giant datasets. One pattern per line:

```
.DS_Store          # macOS clutter
__pycache__/       # Python cache folders
*.log              # any log file
data/raw/          # large data that belongs elsewhere
```

8. Putting a New Project on GitHub

```
# 1. On github.com: click "+" -> "New repository", name it, create it (empty).
# 2. In your project folder:
git init
git add .
git commit -m "First commit"
git remote add origin https://github.com/YOU/project.git
git push -u origin main          # -u links the branch; after this, plain "git push" works
```

> **Commit early, commit often.** Small commits with clear messages ("Add tax column cleaning", not "stuff") make your history readable, and a pushed commit means your work survives a lost or broken laptop.

MSBA Bootcamp · Day 5 · Git & GitHub Basics — companion sheet to Command Line Basics; the daily loop to remember is `status -> add -> commit -> push`.